

Counterfactual Reasoning for Responsibility Attribution in Probabilistic Multi-Agent Systems

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The Problem

Motivation

Trustworthy, accountable MASs need **responsibility-aware agents**. When several agents jointly bring about an outcome, how do we **fairly** attribute responsibility, taking **capability** into account?

Running example

- Two vehicles A_1, A_2 approach an icy junction; each can brake or not.
- If neither brakes $\Rightarrow$ *crash*.
- But slippery road: A_1 fails to stop w.p. 0.2, A_2 w.p. 0.6.

If a crash happens, who is more responsible/to-blame?

Contributions

1. **Backward Counterfactual Responsibility (bCR):**
 - ▶ a Shapley-value degree of responsibility over concurrent stochastic games,
 - ▶ provably *fair* and *consistent*.
2. **Logic:** PATL-SR: PATL + reward + responsibility operators
 - ▶ model checking in PSPACE.
3. Computing **Nash equilibria** in which agents trade off reward against responsibility
 - ▶ also PSPACE.

Backward Counterfactual Responsibility

Model the system as a *concurrent stochastic game* $\mathcal{G}$; agents play *memoryless strategies* $\vec{\sigma}$.

Qualitative bCR

i is responsible for outcome φ under $\vec{\sigma}$ if i belongs to some coalition $J \cup \{i\}$ whose strategies *guarantee* φ , while J alone does *not*.

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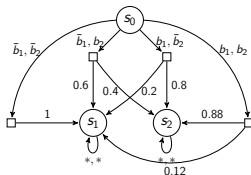
Quantifying it: Shapley value

$$\mathbb{D}^i(\vec{\sigma}, \varphi) = \sum_{J \subseteq \text{Ag} \setminus \{i\}} \frac{|J|!(|\text{Ag}| - |J| - 1)!}{|\text{Ag}|!} (v_{\vec{\sigma}, \varphi}(J \cup \{i\}) - v_{\vec{\sigma}, \varphi}(J))$$

where $v_{\vec{\sigma}, \varphi}(A) = \min_{\vec{\sigma}_{-A}} \text{Prob}(\varphi \mid \vec{\sigma}_A, \vec{\sigma}_{-A})$: other agents play *pessimistically* (trying its hardest to avoid the outcome).

Example

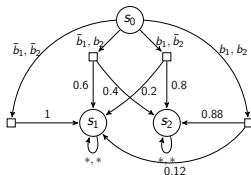
$\varphi := \bigcirc \text{crash}$; both agents choose *not to brake* w.p. 1. Each coalition value $v_{\vec{\sigma}, \varphi}(A)$ is read off the branch where A 's actions are fixed and the rest minimise crash probability:



$$v_{\vec{\sigma}, \varphi}(\{1, 2\})=1, \quad v_{\vec{\sigma}, \varphi}(\{1\})=0.6, \quad v_{\vec{\sigma}, \varphi}(\{2\})=0.2, \quad v_{\vec{\sigma}, \varphi}(\emptyset)=0.12$$

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$$\mathbb{D}^1(\vec{\sigma}, \varphi) = 0.64, \quad \mathbb{D}^2(\vec{\sigma}, \varphi) = 0.24$$

Interpretation

A_1 is *more* responsible: greater chance of stopping (0.8 vs. 0.4) – i.e. greater *capability* to prevent the crash.

Fair and Consistent by Construction

Some important properties, thanks to Shapley value, $\mathbb{D}^i$ satisfies:

- **Efficiency:** $\sum_i \mathbb{D}^i(\vec{\sigma}, \varphi) = \Upsilon(\vec{\sigma}, \varphi)$ (attributable value)
 - ▶ nothing over- or under-assigned.
- **Symmetry & null player:**
 - ▶ equal contribution $\Rightarrow$ equal share;
 - ▶ no contribution $\Rightarrow$ zero share.
- **Monotonicity:** broader outcomes never *decrease* responsibility.
- **(Sub-)additivity:** exact additivity for disjoint, avoidable outcomes.

$$\mathbb{D}^i(\vec{\sigma}, \varphi_1 \vee \varphi_2) = \mathbb{D}^i(\vec{\sigma}, \varphi_1) + \mathbb{D}^i(\vec{\sigma}, \varphi_2)$$

The Logic PATL-SR: Syntax & Verification

PATL-SR extends PATL with **reward** and **responsibility degree** operators:

$$\begin{aligned}\phi &::= a \mid \neg\phi \mid \phi \wedge \phi \mid \langle A \rangle \mathbf{P}_{\bowtie p}[\psi] \mid \langle A \rangle \mathbf{R}_{\bowtie q}[\psi] \\ &\quad \mid \langle A \rangle \mathbf{D}_{\bowtie d}[\mathbf{bCR}_i(\vec{\sigma}, \psi)] \\ \psi &::= \bigcirc\phi \mid \phi \mathbf{U}_{\leq k} \phi\end{aligned}$$

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Verification

Model checking PATL-SR is PSPACE

- no harder than reward-only rPATL.

Responsibility-Aware Strategic Reasoning

Responsibility isn't only measured *after* the fact, agents can *anticipate* it:

Utility

$$u_i^\varphi(\vec{\sigma}) \triangleq \sum_{\rho \in \text{Hist}_{\vec{\sigma}}(s^0)} \wp^i(\rho) - \lambda \cdot \mathbb{D}^i(\vec{\sigma}, \varphi)$$

- Encode strategies as *parameters* $\Rightarrow$ parametric stochastic MAS.
- For reachability/safety φ : u_i^φ is quasi-concave $\Rightarrow$ **NE is computable in PSPACE**.

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Example

Solving the equilibrium gives, braking probabilities $x_1=0.42$, $x_2=0.625$ balance travel-time reward against crash responsibility at equilibrium.

Take-Away

- A **fair, Shapley-based** notion of counterfactual responsibility for probabilistic MAS
 - ▶ capability-sensitive, not just outcome-sensitive.
- PATL-SR lets us *verify* responsibility-aware properties in **PSPACE**.
- Agents can *plan* around anticipated responsibility via **PSPACE**-computable Nash equilibria.

Thank you! 😊